

From pollinosis to digestive allergy.

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Pollinosis is defined as the appearance of respiratory symptoms (rhinoconjunctivitis and/or asthma) as a result of the inhalation of pollen to which the individual is sensitized. Pollen allergy becomes all the more relevant on taking into account that it may be responsible for the development of plant food allergy, or may even constitute the direct cause of esophageal, gastric and/or intestinal inflammation in the context of a digestive allergic process. Pollen can act as a source of allergens that induce primary sensitization in the host as a result of inhalation, with secondary allergy to plant foods containing shared allergens via a cross-reactivity mechanism. The observed pattern of plant food allergy depends on the dietary habits of the population in a given geographical setting, and on the pollination found in that setting. Pollinosis may account for the greater or lesser prevalence of allergy to certain plant foods, and for the severity of the associated reactions. Beyond the digestive tract inflammation that may result from allergy to a given food, pollinosis is also intrinsically able to generate a clinically relevant or irrelevant Th2-mediated inflammatory response at digestive level, and may even give rise to eosinophilic esophagitis. Inter-relation with the airway may also extend to the digestive tract as a consequence of the systemic response that characterizes allergic disease.

[Pollinosis. IV. Which pollens should be tested in allergology practice? Results of determinations of allergy-causing pollens in the Zurich air 1981-1984, with reference to threshold concentrations].

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The most important pollens causing pollinosis in Zurich include hazel (*Corylus*), alder (*Alnus*) and birch (*Betula*) in spring, grasses--including rye--in summer, and mugwort (*Artemisia*) in autumn. Investigations on airborne pollen of Zurich were conducted from 1981 to 1984, and assessed with special reference to threshold concentrations. The results show that sorrel (*Rumex*) and plantain (*Plantago*) also belong to the most important pollens and should therefore be included in the test spectrum for allergological examinations. The pollen of ash (*Fraxinus*), oak (*Quercus*), beech (*Fagus*) and plane tree (*Platanus*) was regularly found in high percentages during these years. Hence the clinical relevance of these tree pollens should not be underestimated, because successful diagnosis and therapy depends mainly on the use of the relevant strain of pollen.

[Allergenic pollens in Spain].

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Allergenic pollens that cause rhinoconjunctivitis and/or asthma are those from trees or plants that pollinate through the air (anemophilic pollination) and not through insects (entomophilic pollination). Although pollen grains would seem to be too large to easily reach the intrapulmonary airways, the relationship between pollen counts and the presence of asthmatic symptoms is only too evident. This is probably because the allergens inducing seasonal asthma are not only found within pollen grains but also outside the grains in particles of less than 10 μ m that are freely found in the atmosphere. The most important pollens producing pollinosis in Spain are those from cypress trees from January-March, birch

trees in April (macizo galaico), *Platanus hispanica* (March-April), grasses and olive trees from April-June, *Parietaria* from April-July and *Chenopodium* and/or *Salsola* from July-September. By geographical areas, the main cause of pollinosis are grasses in the center and north of the peninsula, olive trees in the south (Jaén, Sevilla, Granada, Córdoba) and *Parietaria* in the Mediterranean coast (Barcelona, Murcia, Valencia).

[Cross reactions between pollens and vegetable food allergens].

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The association of food allergies and pollinosis are numerous, implicating tree, grass and weed pollens on one hand and on the other, several plant foods which after ingestion can induce an oral syndrome or more severe reactions such as urticaria, Quincke's edema, asthma and even anaphylactic shock. The molecular basis of cross reactions between pollens and vegetable food allergens is increasingly understood. The principal allergens involved are those of the Bet v 1 family, and profilins found in all pollens as well as in many fruits and vegetables; these two groups of allergens are denatured by high temperatures and by gastric enzymes, in contrast to LTP, which is only found in weeds and some tree pollens. Other molecules can be involved in cross reactions such as Bet v 6 (an isoflavone reductase), 1 beta glucanases and thaumatococcal-like proteins. Inhibition experiments confirmed that the epitopes responsible for primary sensitization come mainly from pollen allergens; the cross-reactive molecular allergen is related to the geographic environment of the patients. The practical aspects of managing these patients are underlined: explanations of co-sensitization, explanations for the lack of efficacy of some extracts, usefulness of a molecular diagnosis obtained either by CAP or microarray, prediction of severe clinical reactions induced by specific molecular allergens and the effectiveness of pollen immunotherapy on the cross-related food allergy.

Allergen component testing - a new era in diagnostics of patients with pollen allergy.

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Introduction: Pollen allergy, also called hay fever or pollinosis, is referred to the most widespread allergic diseases. According to different sources, from 5 to 34% of the population in developed countries are likely to suffer from this condition. Moreover, the majority of patients are young people of working age. The aim: To demonstrate a practical decision of the choice of effective treatment tactics based on component testing in patients with pollen allergy. Materials and methods: 126 patients with allergic rhinitis / conjunctivitis were randomly selected according to their primary visit during the first week in August of the current year. Among them, 53 (42.0%) female and 73 (58.0%) male, aged 22-47 years. General laboratory and instrumental investigations, skin prick test with allergen extracts (Diater, Spain), measurement of total serum and specific IgE by immunoenzymatic assay method using test system "Euroimmun" were performed. Immunofluorescent method ImmunoCap (Thermo Scientific, Uppsala, Swiss) was used to detect specific components of allergens. The material of investigation was blood serum. Results: According to the results of skin prick-test, 50% of patients could receive allergen immunotherapy with two different extracts of allergens "Mixture of Weeds" and "Mixture of Grass". On molecular investigations, it was detected that this combination was not suitable for any patient (20% of individuals had genuine sensitization to allergens of grass pollen, including a marker of cross-reactive molecules, 30% of individuals were sensitized with genuine allergens of *Artemisia* and/or *ambrosia*). Conclusions: Based on skin prick-test and molecular diagnostics, the doctor makes a completely different

decision on the choice of extracts for allergen immunotherapy conduction. Component determination of a sensitized profile and high sensitivity of this method enables to reveal a genuine protein, which is the primary cause of allergy occurrence and administer etiotropic allergen-specific immunotherapy with the allergen, to which sensitization was detected.

Cross-reactivity between grass and corn pollen antigens.

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Analogous reactions of grass and corn pollen extracts in skin tests on patients suffering from pollinosis might suggest an antigenic relationship between grass and corn pollens. This problem was studied using the RAST inhibition test. Tests were performed with cellulose discs labelled with commercial skin test extracts containing grass, rye, wheat, barley, oat and maize pollens. Different mutual inhibitions were measured showing various grades of antigenic relationship. Only grass pollen antigens could strongly inhibit all other antigen-antibody reactions. Thus, we suppose that the investigated grass pollen extract also contains all antigens typical of corn pollen. Therefore, exclusive use of this extract seems to be possible in diagnosis and perhaps therapy of combined grass and corn pollen allergy.

[Case of pollinosis to grass pollens who presented with dyspnea successfully treated by rush immunotherapy].

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We present a 15 years old boy who experienced unusual symptoms for pollen allergy, and successfully treated by rush immunotherapy. The patient started to complain erythema and edema on his face and serous rhinorrhea at 10 years old when going out. He entered baseball team at junior high school, and subsequently experienced choking sensation, dyspnea, face edema, and it was sometimes impossible to continue play. He was diagnosed as bronchial asthma at some hospital, and prescribed many anti-asthma medications including inhaled corticosteroid, which did not take effect. His symptoms deteriorated in summer and ameliorated in winter. When he was 15 years old, he was referred to us by a pediatrician for reassessment of his symptoms. Flow-volume curve was normal, and bronchial provocation test (acetylcholine and histamine), and exercise challenge were negative. IgE antibodies specific to grass pollens were highly positive. We made a diagnosis of pollinosis to grass pollens instead of bronchial asthma. Oral antihistamines and intranasal corticosteroid partially improved his symptoms. We started rush-immunotherapy of grass-pollens (oats and bromegrass), Japanese cedar, and ragweed. His symptoms improved dramatically on the next season of grass pollens.

[Skin sensitization to pollens in Morocco. Multicenter study].

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The objective of this study was to determine the prevalence of skin sensitization to certain pollens in some regions of Morocco. This multicentric study involving 10 centres included a consecutive series of 640 patients consulting for the first time between february and april 1998 for asthma and/or rhinitis

and/or conjunctivitis. Skin prick tests (SPT) were performed with stallerpointes (Stallergenes) and interpreted according to standard procedures. Pollinosis was ascertained when SPT was positive to one or more pollens. Epi-Info was used to analyze the data. SPT were positive in 75.9% and prevalence of sensitization to pollens was equal to 28% (179 cases). It ranked second after house dust mites (63%), olive, 5 grass, cupressus, parietaria and corylus sensitization were encountered in 19.8%, 10.9%, 3.8%, 1.7% and 1.3% respectively. Prevalence of sensitization to pollens didn't vary significantly according to age: 25.4% in females versus 31.1% in males ($p = 0.13$). Prevalence of pollen sensitization was 26.6% in case of asthma versus 30.9% in non asthmatics patients ($p = 0.30$), 30.1% in case of rhinitis versus 16.9% in non rhinitics ($p = 0.009$), 36.4% in case of conjunctivitis versus 22.7% in case without conjunctivitis ($p < 0.0002$). According to the regions, prevalence was significantly high in areas like Meknes (56%), Beni-Mellal (48.1%), Marrakech (43.5%) where olive is abundant in comparison with Oujda (30.8%), Agadir (30.8%), Tanger (27.3%), Rabat (21.8%), Safi (21.3%), Casablanca (11.6%) and El Jadida (10%). The prevalence was up to 84.3% when symptoms were predominant in spring versus 16.8% in other seasons ($p < 0.0001$). Thus, although SPT did not include numerous other pollens, olive pollinosis seems to be prevalent in Morocco, followed by allergy to grass pollen. Such studies need to be carried out in line with pollen count.

Pollen-related allergy to peach and apple: an important role for profilin.

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Birch pollinosis is often accompanied by allergy to fruits such as peach and apple. Bet v I is of major importance as cross-reactive allergen for this combined allergy. We studied a group of patients with combined grass pollinosis and fruit allergy from an area virtually without birch trees. The aim of this study was to investigate the possible involvement of profilin and carbohydrate groups as cross-reactive structures in pollen and fruits. RAST inhibition was performed to measure cross-reactive IgE to pollen and fruits. The presence of IgE against profilin was determined in a RAST with purified grass profilin, and IgE against carbohydrate structures was determined in a RAST with proteinase K-digested grass pollen extract. The biologic activity of IgE in response to profilin was tested by in vitro histamine release and skin prick tests. IgE against fruits was shown to be largely cross-reactive with grass pollen. The majority of the patients had IgE against profilin (12 of 16) and carbohydrate structures (9 of 10). Profilin was shown to have biologic activity, in both histamine release and skin prick tests. Profilin is an important allergen for patients with combined grass pollen/fruit allergy in areas without birch trees.

Aerobiological Monitoring and Metabarcoding of Grass Pollen.

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Grass pollen is one of the leading causes of pollinosis, affecting 10-30% of the world's population. The allergenicity of pollen from different Poaceae species is not the same and is estimated from moderate to high. Aerobiological monitoring is a standard method that allows one to track and predict the dynamics of allergen concentration in the air. Poaceae is a stenopalynous family, and thus grass pollen can usually be identified only at the family level with optical microscopy. Molecular methods, in particular the DNA barcoding technique, can be used to conduct a more accurate analysis of aerobiological samples containing the DNA of various plant species. This study aimed to test the possibility of using the ITS1 and ITS2 nuclear loci for determining the presence of grass pollen from air samples via metabarcoding and to

compare the analysis results with the results of phenological observations. Based on the high-throughput sequencing data, we analyzed the changes in the composition of aerobiological samples taken in the Moscow and Ryazan regions for three years during the period of active flowering of grasses. Ten genera of the Poaceae family were detected in airborne pollen samples. The representation for most of them for ITS1 and ITS2 barcodes was similar. At the same time, in some samples, the presence of specific genera was characterized by only one sequence: either ITS1 or ITS2. Based on the analysis of the abundance of both barcode reads in the samples, the following order could describe the change with time in the dominant species in the air: *Poa*, *Alopecurus*, and *Arrhenatherum* in early mid-June, *Lolium*, *Bromus*, *Dactylis*, and *Briza* in mid-late June, *Phleum*, *Elymus* in late June to early July, and *Calamagrostis* in early mid-July. In most samples, the number of taxa found via metabarcoding analysis was higher compared to that in the phenological observations. The semi-quantitative analysis of high-throughput sequencing data well reflects the abundance of only major grass species at the flowering stage.
